



**SIR SYED UNIVERSITY OF ENGINEERING & TECHNOLOGY**  
**CS&IT DEPARTMENT**  
**PROGRAM BSCS**

**COURSE INFORMATION SHEET**  
**(For Lab Based Course)**

|                                       |                                                                |
|---------------------------------------|----------------------------------------------------------------|
| <b>Session:</b>                       | Fall-2024                                                      |
| <b>Course Title:</b>                  | Web Engineering                                                |
| <b>Course Code:</b>                   | CS-311L                                                        |
| <b>Credit Hours:</b>                  | 0+1                                                            |
| <b>Semester:</b>                      | 5 <sup>th</sup>                                                |
| <b>Pre-Requisites:</b>                | PROGRAMMING FUNDAMENTALS (CS-116L)                             |
| <b>Instructor Name:</b>               | Mr. Muhammad Rahil Usman                                       |
| <b>Email and Contact Information:</b> | <a href="mailto:rausman@ssuet.edu.pk">rausman@ssuet.edu.pk</a> |
| <b>WhatsApp Group</b>                 | WebEngineering2024                                             |
| <b>Office Hours:</b>                  | <b>Monday to Friday (09:00 AM to 05:00 PM)</b><br>✓            |
| <b>Mode of Teaching:</b>              | <b>Synchronous/Asynchronous/ Hybrid/Blended</b>                |

**Course Objectives:**

Upon successful completion of the course, the student will demonstrate the ability to:

- Understand the major areas and challenges of web programming.
- Distinguish web-related technologies.
- Be able to build ANY website you want
- Build fully-fledged websites and web apps for your startup or business
- Understand the latest frameworks and technologies, including Bootstrap 4, Mongo DB, NodeJS, Express.

**Course Description:**

Web Design: HTML 5, CSS 3. Responsive Web Design: Bootstrap 5. Front-End Web Development: JavaScript ES6, DOM Manipulation, JQuery, Bash Command Line, Git, GitHub and Version Control. Backend Web Development: Node.js, NPM, Express.js, REST, APIs. Database: MongoDB. Deployment: with GitHub Pages, Heroku, and Mongo DB Atlas



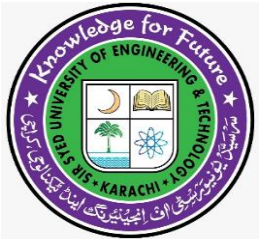
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**COURSE LEARNING OUTCOMES (CLOs) and its mapping with Program Learning Outcomes (PLOs):**

| CLO No. | Course Learning Outcomes (CLOs)                                                                                                                          | PLOs | Bloom's Taxonomy |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------|------|------------------|
| 1       | <b>Describe</b><br>Web standards, web applications Modern protocols, programming concepts and control structures used in web programming.                | 1    | Remember         |
| 2       | <b>Explain</b><br>Programs having HTML,CSS,JavaScript protocols standards.                                                                               | 1    | Understand       |
| 3       | <b>Apply</b><br>Apply knowledge and understanding to solve interactive web development problems and project their solutions under specific requirements. | 3    | Apply            |

**GRADING POLICY:**

| Assessment Tools   | Percentage  | Marks     |
|--------------------|-------------|-----------|
| Lab Manual         | 30%         | 15        |
| Viva/Demonstration | 30%         | 15        |
| Lab Exam/Project   | 40%         | 20        |
| <b>TOTAL</b>       | <b>100%</b> | <b>50</b> |



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## Recommended Book:

- Jon Duckett HTML and CSS Design and Build Websites
- **FREEMAN, ADAM THE DEFINITIVE GUIDE TO HTML5:**
- Ladwa, Hanumanth HTML, CSS, and JavaScript
- Terry Ann Felke-Morris, Ed.D Web Design HTML5 & CSS
- A Smarter Way to Learn JavaScript

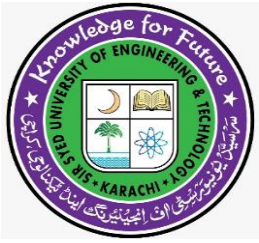
## Reference Books:

- Emilia Mendes and Nile Mosley, Springer Verlag Web Engineering, , 2010.
- Roger S. Pressman, McGraw Hill, Web Engineering: A Practitioners' Approach, 2008.
- Dynamic HTML: The Definitive Reference: A Comprehensive Resource for XHTML, CSS, DOM, JavaScript O'Reilly Media 3rd Edition, 2007.
- David Flanagan. O'Reilly Media JavaScript: The Definitive Guide, 8th Edition,.
- Dreamtech Press Black Book Web Technologies, ,
- P.J. Deitel & H.M. Deitel Pearson. Internet and World Wide Web How to program,

## LAB PLAN

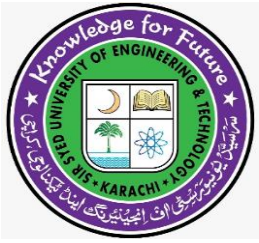
**Course Title: Web Engineering**  
**Course Code: CS-311**

| Week # |                                | Topics                                                                                                                                                                                                                                                  | Required Reading |
|--------|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1      | 14-10-2024<br>to<br>18-10-2024 | <b>Client-side technologies: HTML-I</b> <ul style="list-style-type: none"><li>• The Anatomy of an HTML Tag</li><li>• What we're building - HTML Personal Site</li><li>• What is The HTML Boilerplate?</li><li>• How to Structure Text in HTML</li></ul> | Lab Manual       |



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|   |                                |                                                                                                                                                                                                                                                                                                                                                                                |                   |
|---|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
|   |                                | <ul style="list-style-type: none"> <li>HTML Lists</li> <li>HTML Image Elements</li> <li>HTML Links and Anchor Tags</li> </ul>                                                                                                                                                                                                                                                  |                   |
| 2 | 21-10-2024<br>to<br>25-10-2024 | <b>Client-side technologies: HTML-II</b> <ul style="list-style-type: none"> <li>HTML Tables</li> <li>Using HTML Tables for Layout</li> <li>HTML Tables Code Challenge</li> <li>How to Type Emojis</li> <li>HTML Forms</li> </ul>                                                                                                                                               |                   |
| 3 | 28-10-2024<br>to<br>01-11-2024 | <b>Client-side technologies: CSS - I</b> <ul style="list-style-type: none"> <li>Introduction to CSS</li> <li>The Anatomy of CSS Syntax</li> <li>Inline CSS, Internal CSS and External CSS</li> <li>CSS Selectors</li> <li>Classes vs. Ids</li> </ul>                                                                                                                           |                   |
| 4 | 04-11-2024<br>to<br>08-11-2024 | <b>Client-side technologies: CSS - II</b> <ul style="list-style-type: none"> <li>What Are Favicons?</li> <li>Divs</li> <li>The Box Model of Website Styling</li> <li>CSS Display Property</li> <li>CSS Static, Relative and Absolute positioning</li> <li>CSS Typography</li> </ul>                                                                                            | <b>Quiz</b>       |
| 5 | 11-11-2024<br>to<br>15-11-2024 | <b>Client-side technologies: Bootstrap</b><br>What is Bootstrap? <ul style="list-style-type: none"> <li>Installing Bootstrap</li> <li>Web Design - Wire framing</li> <li>The Bootstrap Navigation Bar</li> <li>Bootstrap Grid Layout System</li> <li>Adding Grid Layouts to Our Website</li> <li>Bootstrap Containers</li> <li>Bootstrap Buttons &amp; Font Awesome</li> </ul> | <b>Assignment</b> |
| 6 |                                | <ul style="list-style-type: none"> <li><b>Client-side technologies: Javascript - I</b></li> <li>Introduction to JavaScript</li> <li>JavaScript Alerts - Adding Behavior to Websites</li> <li>Data Types</li> <li>JavaScript Variables</li> </ul>                                                                                                                               |                   |



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|    |                                                                  |                                                                                                                                                                                                                                                                                                                                                                               |                   |
|----|------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
|    | 18-11-2024<br>to<br>22-11-2024                                   | <ul style="list-style-type: none"> <li>String Functions</li> <li>Basic Arithmetic and the Modulo Operator in JavaScript</li> <li>Increment and Decrement Expressions</li> <li>Control Statements: Using If-Else Conditionals &amp; Logic</li> <li>Control Statements: While Loops</li> <li>Control Statements: For Loops</li> <li>JavaScript Objects and Functions</li> </ul> |                   |
| 7  | 25-11-2024<br>to<br>29-11-2024                                   | <b>Client-side technologies: Javascript - II</b>                                                                                                                                                                                                                                                                                                                              |                   |
| 8  | <b>MID TERM EXAMINATION</b><br><b>(02-12-2024 to 07-12-2024)</b> |                                                                                                                                                                                                                                                                                                                                                                               |                   |
| 9  | 09-12-2024<br>to<br>13-12-2024                                   | <b>Client-side technologies: jQuery and JSON</b>                                                                                                                                                                                                                                                                                                                              |                   |
| 10 | 16-12-2024<br>to<br>20-12-2024                                   | <b>Client-side technologies: AJAX and React JS</b> <ul style="list-style-type: none"> <li>Learn the purpose of Asynchronous JavaScript And XML (AJAX)</li> </ul>                                                                                                                                                                                                              |                   |
| 11 | 23-12-2024<br>to<br>27-12-2024                                   | <b>Server-side technologies: C#, WebAPI Basics</b>                                                                                                                                                                                                                                                                                                                            |                   |
| 12 | 30-12-2024<br>to<br>03-01-2025                                   | <b>Server-side technologies: LINQ queries, MVC Framework, C# Data First approach</b>                                                                                                                                                                                                                                                                                          | <b>Quiz</b>       |
| 13 | 06-01-2025<br>to<br>10-01-2025                                   | <b>Server-side technologies: C# Data First Approach with Web APIs – Web API Consumption</b>                                                                                                                                                                                                                                                                                   | <b>Assignment</b> |
| 14 | 13-01-2025<br>to<br>17-01-2025                                   | <b>Deployment, Operation &amp; Maintenance of Web Applications</b>                                                                                                                                                                                                                                                                                                            |                   |
| 15 |                                                                  | <b>Final Exam</b>                                                                                                                                                                                                                                                                                                                                                             |                   |